MIL-P-48099A (AR)
30 January 1978

SUPERSEDING
MIL-P-48099 (MU)
13 March 1973

MILITARY SPECIFICATION

PROPELLANT, M10
(FOR USE IN SPECIAL APPLICATIONS)

This specification is approved for use by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers M10 Propellant for use in the 155MM, M483A1 projectile and 8 inch, M509E1 projectile (see 6.3).

1.2 Classification. The propellant shall be M10 flake.

2. APPLICABLE DOCUMENTS.

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-652 - Propellants, Solid, for Cannons, Requirements and Packing

FSC: 1376

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:13Z
Check the source to verify that this is the current version before use.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The propellant shall be M10 Flake, conforming to the requirements of MIL-STD-652 with the following exceptions:

TABLE I

Requirements	Percentage
Nitrocellulose	98.00 ± 1.00
Potassium Sulfate	1.00 ± 0.30
Diphenylamine	1.00 ± 0.30
Graphite (glaze added) Max.	0.25
Total volatiles, Max.	2.00
Moisture, Max.	1.20
Moisture, Min.	0.50

3.1.1 Grain characteristics.

3.1.1.1 Form of grain. The propellant shall be in the form of irregular circular flakes.

3.1.1.2 Grain dimensions. The dimensions of the grain shall be as follows:

Diameter (Advisory)	0.035 inch
Thickness (Web) Advisory	0.011 inch

3.1.2 Heat test 134.5° degrees centigrade.(c). The propellant shall not cause complete fading of methyl violet test paper to a salmon pink color in less than 50 minutes, when tested as specified in 4.4.1.

3.1.3 Closed Bomb (Relative Quickness and Relative Force). The M10 propellant shall produce a relative quickness and a relative force of 100 ± 3.0 percent (%) when tested as specified in 4.4.2.

3.1.4 Data requirements. The contractor shall generate data, in accordance with the requirements of the data item descriptions cited in 4.4.2.1, documenting specific laboratory and Proving Ground test data.

3.1.5 Workmanship. The best commercial practices shall be used in the formulation of propellant furnished under this

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:13Z
Check the source to verify that this is the current version before use.

specification, and all other applicable documents. The propellant and its standard ingredients shall be protected from the action of direct sunlight and acid fumes. Unless otherwise specified, no reground propellant or nitrocellulose shall be used in the manufacture of propellant procured under this specification.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. Quality Conformance Inspection shall be conducted on this item.

4.3 Quality conformance inspection.

4.3.1 Inspection lot formation. Inspection lots shall comply with the lot formation provision of MIL-A-48078.

4.3.2 Examination and testing.

4.3.2.1 Sampling for laboratory testing (see 3.1). Each lot of propellant shall be sampled and tested for composition, form dimensions and chemical/physical properties in accordance with MIL-STD-652. The contractor shall submit a Propellant Description Sheet in report form and in computer-compatible format (keypunch cards) in accordance with Data Item Description DI-T-4904, giving the detailed test results (see 6.1.2).

4.4 Test methods and procedures.

4.4.1 Chemical and physical tests. The chemical and physical tests shall be conducted in accordance with Standard MIL-STD-286.

TABLE II
MIL-STD-286 Test Methods

Chemical and Physical Test	Method Number
Nitrocellulose	209.2
Potassium Sulfate	310.4 or 316.1
Diphenylamine	201.1 or 226.2
Graphite	308.1 or 309.1
Total Volatiles	103.3 or 103.5
Moisture	102.1 or 103.5
134°C Heat Test	404.1
Dimension of Grains	504.1

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:13Z
Check the source to verify that this is the current version before use.

4.4.2 Closed Bomb. The relative quickness and relative force shall be determined in accordance with method 801.1 of standard MIL-STD-286 with the following exceptions: Propellant lot DZ-DUP-1-25 (see 6.4), with a thickness of 0.011 inch shall be used as the reference standard. In addition, the following test parameters are specified:

Volume of Bomb - 200 cc Nominal
Loading Density (including ignition material) - 0.1 gm/cc
Sequence of Firing - 3 shots of standard and
3 shots of test propellant
fired alternately.

5. PACKAGING

5.1 Packing and marking. Packing and marking shall be in accordance with MIL-STD-652, except that all propellant designated for use as references shall be packed Level A (see 6.4).

6. NOTES

6.1 Ordering data. Ordering data shall be in accordance with MIL-A-48078 (PA) with the following exceptions.

6.1.1 Procurement requirements. Procurement requirements of MIL-A-48078 shall apply except for First Article Sample.

6.1.2 Contract data requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data Requirement	Applicable Data Item Description
4.3.2.1	Propellant Description Sheet	DI-T-4904
4.3.2.1	Keypunched Cards	DI-T-4904

One copy each of the Propellant Description Sheet:

Commander
U.S. Army Armament Research and Development Command
ATTN: DRDAR-QAR
DRDAR-LCA-G
Dover, New Jersey 07801

One copy of each set of keypunched cards shall be forwarded, ATTN: DRDAR-QAR-R. (Copies of data item descriptions required

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:13Z
Check the source to verify that this is the current version before use.

by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

6.2 Surveillance sample. Samples of each lot shall be forwarded to US Army R&D Command, Dover, New Jersey, ATTN: DRDAR-LCE-MP for the propellant surveillance test program in accordance with MIL-STD-652. A 5 gram reference sample shall be forwarded to DRDAR-QAR-R for stockpile evaluation, whenever, a reference lot is selected.

6.3 Intended Use. The propellant in this specification is intended for use as an expelling charge.

6.4 Reference propellant. The reference sample is controlled by Radford Army Ammunition Plant, ATTN: SARRA-Q, Radford, Virginia 07424.

Custodian:
Army-AR

Preparing Activity:
Army-AR

Project Number: 1376-A092

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:13Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:13Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.
NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:13Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY
US ARMY ARMAMENT RESEARCH AND
DEVELOPMENT COMMAND
ATTN: DRDAR-TST-S
DOVER, NJ 07801

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

POSTAGE AND FEES PAID

DEPARTMENT OF THE ARMY

DOD - 314

U.S. MAIL

Commander
US Army Armament Research & Development Command
ATTN: DRDAR- QAA
Dover, NJ 07801

FOLD

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:13Z
Check the source to verify that this is the currentversion before use.